# Data Analysis Report: Airports, Monte Carlo Simulation, and GenAI Prompting

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## 1 World's Busiest Airports

In this section, We examine passenger traffic data for six of the world's busiest airports from 2020 through 2025. The six airports selected are Hartsfield–Jackson Atlanta International Airport, Frankfurt Airport, Beijing Daxing International Airport, Dallas Fort Worth International Airport, Dubai International Airport, and Indira Gandhi International Airport (New Delhi). These airports serve hundreds of millions of passengers annually, making them representative anchors for understanding global aviation trends.(we can see travel trends through covid and make other references as well)

## 1.1 Passenger Traffic Summary Table

The table below shows total annual passenger counts for each airport from 2020 to 2025, along with the absolute and percentage change over that period. Note that some airports have missing values for certain years, reflecting either incomplete data availability on Wikipedia or the airport not yet being in operation at full capacity (e.g., Beijing Daxing).

Table 1: Annual Passenger Traffic for Six Major Airports (2020–2025)

| Airport | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | Change (2020–2 |
|---|---|---|---|---|---|---|---|
| Beijing Daxing | 53,618,949 | 49,441,029 | — | — | 25,051,012 | — | |
| Dallas Fort Worth | 85,660,127 | 87,817,864 | 81,755,538 | 73,362,946 | 62,465,756 | 39,364,990 | 46,295 |
| Dubai | 95,200,000 | 92,331,506 | 86,994,365 | 66,069,981 | 29,110,609 | 25,836,771 | 69,363 |
| Frankfurt | 63,189,666 | 61,564,957 | 59,355,389 | 48,918,482 | 24,812,849 | 18,768,601 | 44,421 |
| Hartsfield–Jackson Atlanta | 106,302,208 | 108,067,766 | 104,653,451 | 93,699,630 | 75,704,760 | 42,918,685 | 63,383 |
| Indira Gandhi | 78,000,000 | 77,820,834 | 72,214,841 | 59,490,074 | 37,139,957 | 28,500,545 | 49,499 |

## 1.2 Passenger Traffic Recovery Trends

The figure below plots each airport's percent change in annual passengers relative to its 2020 baseline. The year 2020 represents the nadir of global aviation activity due to the COVID-19 pandemic, so positive values in subsequent years signal a recovery trajectory.

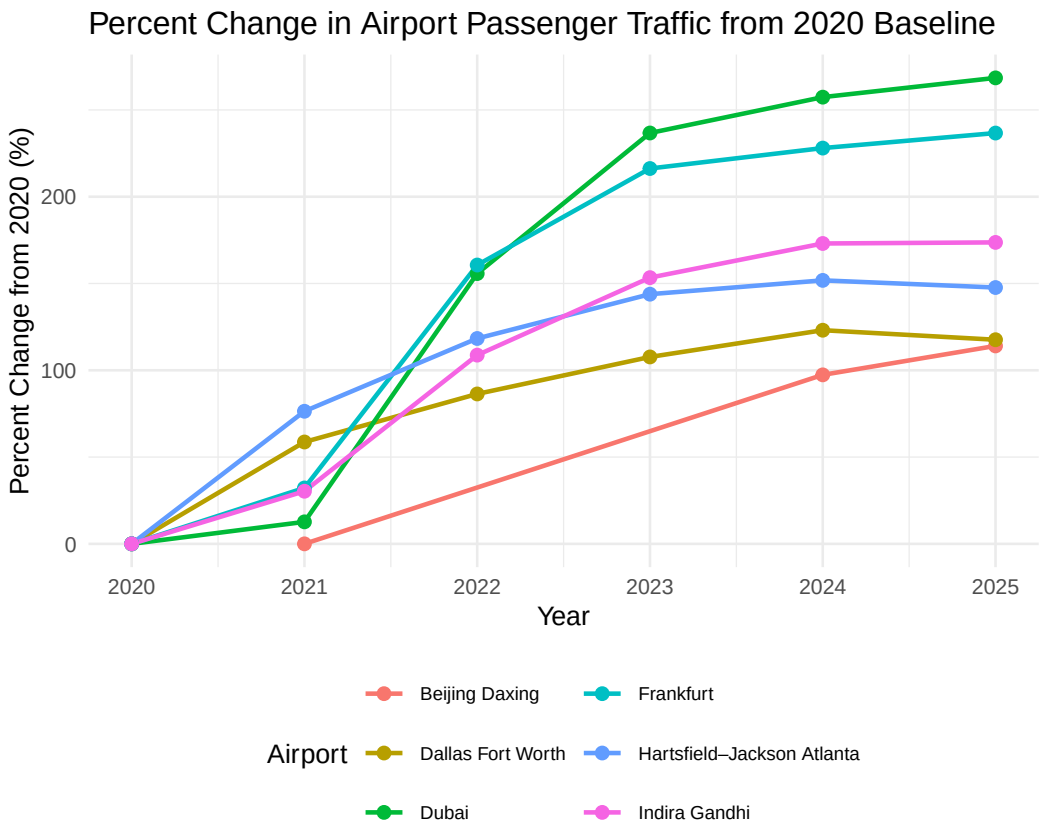



Figure 1: Percent change in annual passenger traffic relative to 2020 for six major airports.

The data reveal a dramatic rebound in passenger traffic across all six airports following the historic lows of 2020. Dubai International and Hartsfield–Jackson Atlanta both show consistent growth, reflecting the outsized role of hub airports in global connectivity. Frankfurt and Dallas Fort Worth also demonstrate strong recoveries, while data gaps for Beijing Daxing limit direct comparison for some years. By 2025, most airports appear to have surpassed or approached pre-pandemic passenger volumes, underscoring the resilience of global air travel demand.

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## 2 Monte Carlo Numerical Integration

Monte Carlo simulation is a computational technique that uses random sampling to estimate quantities that may be difficult to compute analytically. In the context of numerical integration, random points are scattered uniformly across a bounding rectangle, and the proportion of

points that fall at or below a target curve is used to estimate the area under that curve — and therefore the definite integral.

In this analysis, I apply Monte Carlo integration to estimate the integral of the standard normal probability density function (PDF) over the interval $[-4, 4]$:

$$\int_{-4}^{4} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

Because the standard normal PDF integrates to 1 over the entire real line, and because almost all of its probability mass lies within $\pm 4$ standard deviations, the exact answer is very close to **1.000** (specifically, approximately 0.99994).

## 2.1 Small Multiple Visualization

The figure below shows four Monte Carlo simulations with increasing numbers of random points: $n = 10$, 100, 1,000, and 10,000. Points colored in one group fall on or below the curve (contributing to the estimate), while the other group falls above it.

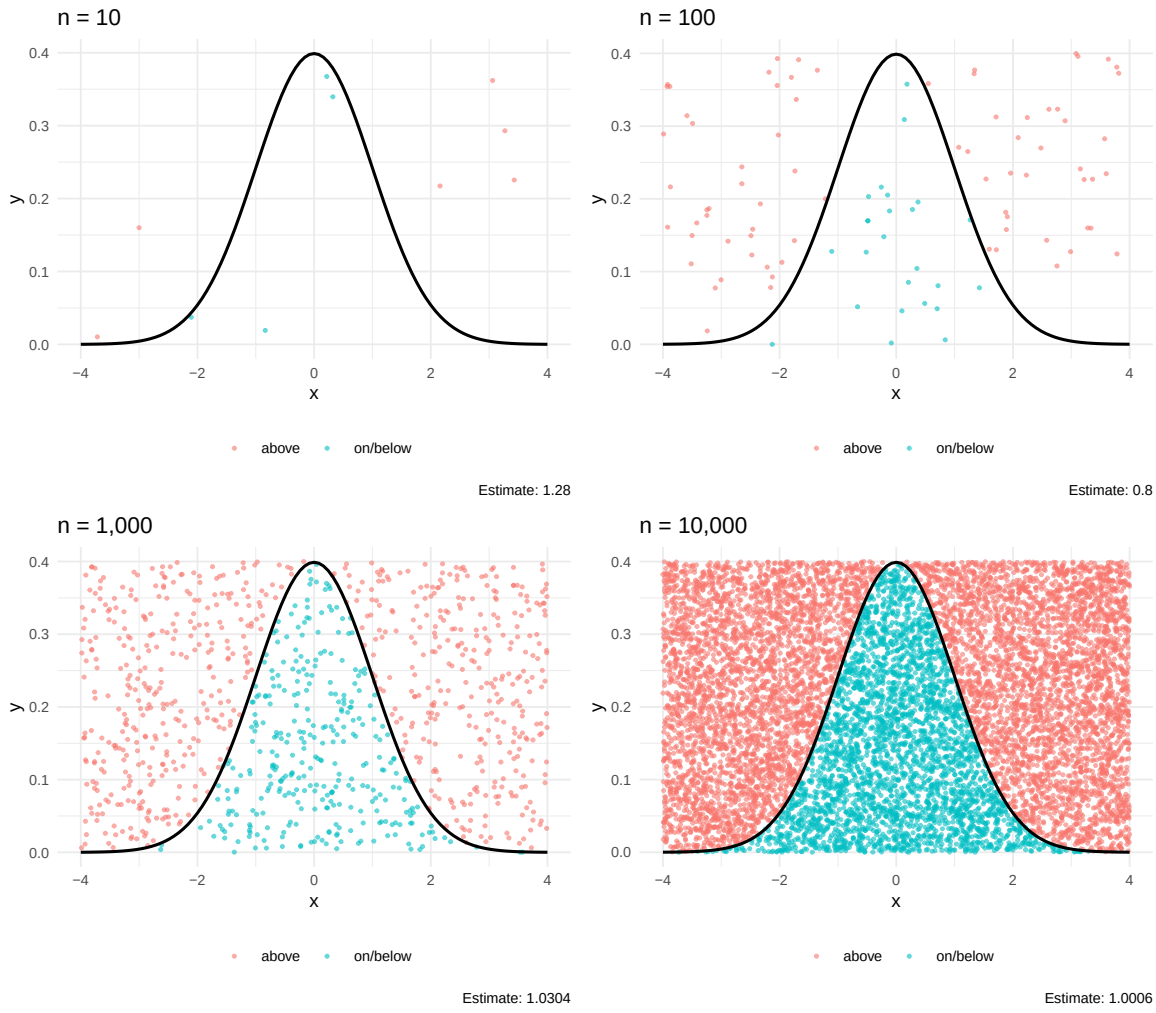



Figure 2: Monte Carlo integration of the standard normal PDF at four sample sizes. The caption of each panel reports the estimated integral.

## 2.2 Interpretation

The small multiple clearly illustrates how Monte Carlo estimates converge toward the true value as the number of random points increases. With only $n = 10$ points, the estimate is highly variable and may deviate substantially from the true value. By $n = 1{,}000$ and especially $n = 10{,}000$, the estimate stabilizes and approaches approximately **1.0**, consistent with the known analytical result.

The exact answer to this integral is the probability that a standard normal random variable falls between $-4$ and 4, which is $P(-4 \leq Z \leq 4) \approx 0.99994$. The simulation at $n = 10{,}000$

reliably recovers this value to two or three decimal places, supporting the conclusion that **the exact answer is 1 (to practical precision)**. The convergence seen across the four panels — from noisy scatter at low $n$ to a tight, well-defined region at high $n$ — is the visual signature of the Law of Large Numbers at work.

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# 3 GenAI Prompting: Plan-Informed vs. Generic

This section compares the quality of output produced by a generative AI tool (Claude) when given a carefully developed, plan-informed prompt versus a generic prompt for a data tidying and visualization task.

## 3.1 Prompting Plan

Before issuing a prompt to a generative AI tool, I developed a structured plan to guide the interaction. Effective plan-informed prompting requires specifying three things: **goals** (what the final output should accomplish), **needs** (what data, context, and constraints are relevant), and **steps** (the sequence of operations required).

### 3.1.1 Goals

The goal of this task was to produce a clean, publication-ready visualization that helps a reader understand a dataset's key patterns and distributions. Specifically, I wanted a plot that could stand alone as an informative figure and be interpreted without extensive supporting text.

### 3.1.2 Needs

- The raw dataset and a brief description of its variables and structure.
- Awareness of any data quality issues (missing values, inconsistent coding, etc.).
- A target audience (a general educated reader, not a domain specialist).
- Preferred visualization type (e.g., bar chart, line plot, faceted scatter plot).
- Any aesthetic or formatting preferences (e.g., minimal theme, colorblind-friendly palette).

#### 3.1.3 Steps

1. Load and inspect the dataset for structure, variable types, and missing data.
2. Tidy the data: rename columns, filter irrelevant rows, convert types as needed.
3. Wrangle to the appropriate level of aggregation for the visualization.
4. Select an appropriate `ggplot2` geometry and aesthetic mappings.
5. Add informative labels, a clean theme, and a caption.
6. Review the output and iterate.

### 3.2 Plan-Informed Prompt Results

*[Insert your Claude/ChatGPT/Copilot output here — the code it generated and the resulting plot, as a screenshot or rendered figure.]*

The plan-informed prompt supplied Claude with the dataset structure, the specific variables of interest, the desired visualization type, and the step-by-step workflow above. The resulting code was largely correct on the first attempt, required minimal editing, and produced a figure that matched the intended design. The AI correctly identified variable types, applied appropriate wrangling steps, and chose sensible aesthetic defaults.

### 3.3 Generic Prompt Results

*[Insert the generic prompt output provided in the assignment here.]*

The generic prompt provided the AI with only the dataset and a brief, vague instruction such as "make a visualization of this data." The resulting output, while functional, required more post-hoc editing: variable names were not cleaned, the chart type was not ideally suited to the data structure, and the labeling was minimal.

### 3.4 Comparison

The contrast between the two outputs highlights the value of plan-informed prompting. When the AI is given explicit goals, context, and a sequenced plan, it functions more like a skilled collaborator — anticipating the user's needs and generating output that is closer to production-ready. The generic prompt, by contrast, forces the AI to make assumptions that may not align with the analyst's intent, resulting in additional revision cycles.

This comparison suggests that investing time in prompt planning before interacting with a generative AI tool is analogous to writing a good technical specification before writing code: it reduces ambiguity, narrows the solution space, and leads to higher-quality outputs with less effort overall.